Locking Away Prosperity? Evaluating Economic Impact of the Key to NYC Program

1 Introduction

The onset of the COVID-19 pandemic triggered a global wave of non-pharmaceutical interventions (NPIs), as governments acted swiftly to limit viral transmission. Early research focused heavily on the public health rationale behind these policies—particularly the importance of acting early to contain exponential spread. For example, Friedson et al. [2021] and Yang [2021] evaluate the effects of lockdowns in California and Wuhan, emphasizing the critical role of timing. Bayat et al. [2020] estimate that New York City could have reduced COVID-19 deaths by over 80% had its first lockdown been enacted just ten days earlier.

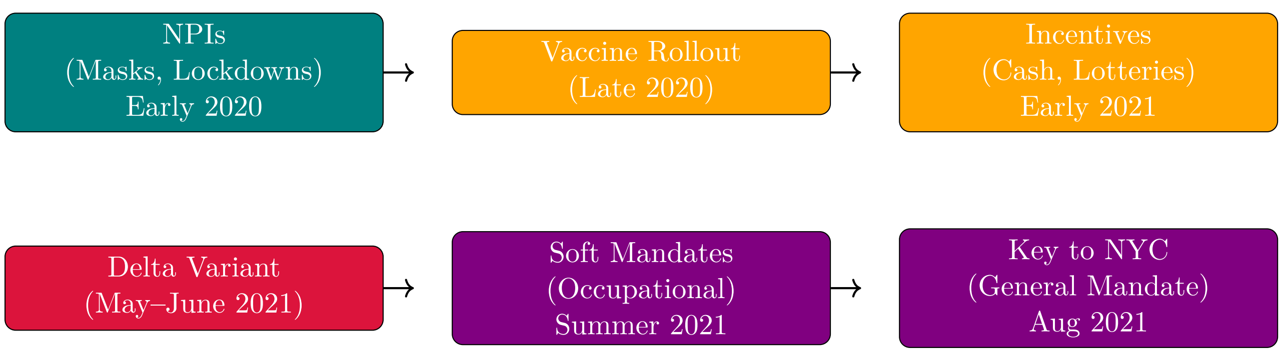
Over time, attention turned to the economic consequences of these policies. While many NPIs helped flatten the curve, they also generated concerns about business closures, layoffs, and long-term economic scarring. Studies such as Gibson and Sun [2023] and Lee and Yang [2022] document the labor market costs of early lockdowns, while Ke and Hsiao [2022] use a panel data approach to examine the short-run economic toll of Hubei's province-wide shutdown. These papers illustrate the inherent tradeoffs between public health benefits and economic costs embedded in pandemic interventions.

Vaccine mandates, unlike lockdowns or school closures, have received comparatively less economic scrutiny, especially regarding their effects on labor markets. These policies aimed to reduce transmission without broadly restricting business operations or movement. Nevertheless, critics warned mandates could hamper fragile economic recoveries by alienating customers and straining staffing in consumer-facing sectors like restaurants and hospitality. Proponents countered that mandates would enhance public safety and bolster consumer confidence. New York City's "Key to NYC" program—the first large-scale vaccine mandate for indoor commercial activity in the U.S., implemented in August 2021—sparked national debate along these lines [Towey, 2021, Rogers, 2021, Bardosh et al., 2022]. This controversy highlights the empirical question: *Did the mandate harm employment in NYC's restaurant sector?*

This paper treats the Key to NYC program as a natural experiment, exploiting its sudden rollout to identify its causal impact on full-service restaurant employment—the segment most directly affected. While prior research has focused on mandates' effects on health outcomes and vaccination rates [Cohn et al., 2022, Acharya and Dhakal, 2021, Lang et al., 2023], few have rigorously examined labor market consequences in the places most likely to be affected by them. Estimating these effects is complicated by overlapping pandemic trends and economic shifts. While prior research has evaluated the Key to NYC program's effects on vaccination uptake and public health, this study is among the first to isolate its economic impact on the full-service restaurant sector—the area arguably most exposed to the policy's constraints.

Randomized experiments are infeasible and unethical for assigning vaccine mandates to cities, so we must estimate counterfactuals using observational data. These counterfactuals can never be directly observed, but they can be approximated. I construct a synthetic control for the year-over-year growth rate of full-service restaurant employment in New York City using seasonally adjusted MSA-level data from the Federal Reserve Bank of St. Louis. The dataset spans January 1991 to August 2022, allowing for a long pre-treatment window. This is particularly valuable given the properties of the synthetic control method, which is most reliable in the fixed-N, large-T setting. A longer pre-treatment period helps improve the stability and credibility of counterfactual estimates, provided there are no structural breaks affecting only the treated unit. There are 29 donor units that FRED reports on to construct the counterfactual from.[1] As a robustness check, I also examine employment in the broader Leisure and Hospitality sector, which may capture indirect spillover effects or broader sectoral shocks correlated with restaurant trends.

2 Policy Background



Timeline of major pandemic policy phases

Building on early reliance on non-pharmaceutical interventions (NPIs) such as lockdowns and masking, the arrival of vaccines in late 2020 ushered in a new phase of pandemic governance. NPIs had aimed to reduce transmission by limiting contact or promoting respiratory barriers. Mask mandates were widely adopted due to their low cost and direct mechanism of action [Hansen and Mano, 2023, Chernozhukov et al., 2021, Mitze et al., 2020], while lockdowns—though effective—provoked political and economic backlash due to their disruptive nature.

[1] For example, one such series is 'All Employees: Leisure and Hospitality: Full-Service Restaurants in District of Columbia (SMU11000007072251101SA)' or 'SMU08197407072251101SA' for Denver-Aurora-Centennial, CO (MSA). These cities did not implement similar mandates during the same period, and thus serve as a "donor pool" for constructing the counterfactual.

With vaccines came the promise of exit, and initial uptake was high, particularly among older adults and healthcare workers. However, as eligibility expanded in early 2021, demand slowed amid misinformation, mistrust, and logistical barriers. In response, policymakers deployed incentive-based strategies to encourage vaccination [Barber and West, 2022, Robertson et al., 2021]. These included direct cash payments, vaccine lotteries (e.g., Ohio's "Vax-a-Million" [Fuller et al., 2022]), and smaller symbolic giveaways—ranging from donuts and beer to cannabis and firearms. One paper famously dubbed this the era of "donuts, drugs, booze, and guns," reflecting the extraordinary lengths to which governments went to boost uptake through rewards rather than requirements [Tinari and Riva, 2021].

Despite their creativity, most incentives had modest or short-lived effects [Saban et al., 2021, Walkey et al., 2021, Khazanov et al., 2023]. Many were poorly targeted, administratively complex, or insufficient to overcome deeper resistance. The emergence of the highly transmissible Delta variant in mid-2021 raised the stakes. Initially identified in India, Delta reached the United States by May and accounted for the majority of new cases by June. Unlike in 2020, officials now had vaccines at scale—but not enough vaccinated people.

Public health responses escalated. Soft mandates emerged, targeting workers in high-risk occupations such as healthcare and education. These policies were not general mandates but did make vaccination a condition of continued employment [Khunti et al., 2021]. Over the summer, local governments expanded this approach. By August 2021, New York City introduced the Key to NYC program—one of the first large-scale vaccine mandates in the U.S. applied to the general public. Unlike prior efforts that relied on encouragement, Key to NYC made access to indoor dining, fitness, and entertainment conditional on vaccination status. It represented a new policy mechanism: exclusion rather than incentive.

References

Binod Acharya and Chandra Dhakal. Implementation of state vaccine incentive lottery programs and uptake of covid-19 vaccinations in the united states. *JAMA Network Open*, 4(12):e2138238–e2138238, 12 2021. ISSN 2574-3805. doi: 10.1001/jamanetworkopen.2021.38238.

Andrew Barber and Jeremy West. Conditional cash lotteries increase covid-19 vaccination rates. *Journal of health economics*, 81:102578, 2022.

Kevin Bardosh, Alex de Figueiredo, Rachel Gur-Arie, Euzebiusz Jamrozik, James Doidge, Trudo Lemmens, Salmaan Keshavjee, Janice E. Graham, and Stefan Baral. The unintended consequences of covid-19 vaccine policy: why mandates, passports and restrictions may cause more harm than good. *BMJ Global Health*, 7(5):e008684, 2022. doi: 10.1136/bmjgh-2022-008684. URL https://doi.org/10.1136/bmjgh-2022-008684.

Niloofar Bayat, Cody Morrin, Yuheng Wang, and Vishal Misra. Synthetic control, synthetic interventions, and covid-19 spread: Exploring the impact of lockdown measures and herd immunity, 2020. URL https://arxiv.org/abs/2009.09987.

Victor Chernozhukov, Hiroyuki Kasahara, and Paul Schrimpf. Causal impact of masks, policies, behavior on early covid-19 pandemic in the u.s. *Journal of Econometrics*, 220(1): 23–62, 2021. doi: 10.1016/j.jeconom.2020.09.003.

Ezra Cohn, Michael Chimowitz, Theodore Long, Jay K Varma, and Dave A Chokshi. The effect of a proof-of-vaccination requirement, incentive payments, and employer-based mandates on covid-19 vaccination rates in new york city: a synthetic-control analysis. *The Lancet Public Health*, 7(9):e754–e762, 2022.

Andrew I. Friedson, Drew McNichols, Joseph J. Sabia, and Dhaval Dave. Shelter-in-place orders and public health: Evidence from california during the covid-19 pandemic. *Journal of Policy Analysis and Management*, 40(1):258–283, 2021. doi: 10.1002/pam.22267.

Sam Fuller, Sara Kazemian, Carlos Algara, and Daniel J. Simmons. Assessing the effectiveness of covid-19 vaccine lotteries: A cross-state synthetic control methods approach. *PLOS ONE*, 17(9):1–18, 09 2022. doi: 10.1371/journal.pone.0274374.

John Gibson and Xiaojin Sun. A synthetic control analysis of us state level covid-19 stay-at-home orders on new jobless claims. *Journal of Economics and Finance*, 47(1):1–14, 2023.

Niels-Jakob H. Hansen and Rui C. Mano. Mask mandates save lives. *Journal of Health Economics*, 88:102721, 2023. ISSN 0167-6296. doi: https://doi.org/10.1016/j.jhealeco.2022.102721. URL https://www.sciencedirect.com/science/article/pii/S0167629622001357.

Xiao Ke and Cheng Hsiao. Economic impact of the most drastic lockdown during covid-19 pandemic—the experience of hubei, china. *Journal of Applied Econometrics*, 37(1):187–209, 2022. doi: 10.1002/jae.2871.

Gabriela K. Khazanov, Rebecca Stewart, Matteo F. Pieri, Candice Huang, Christopher T. Robertson, K. Aleks Schaefer, Hansoo Ko, and Jessica Fishman. The effectiveness of financial incentives for covid-19 vaccination: A systematic review. *Preventive Medicine*, 172:107538, 2023. doi: 10.1016/j.ypmed.2023.107538.

Kamlesh Khunti, Atiya Kamal, Manish Pareek, and Amanda Griffiths. Should vaccination for healthcare workers be mandatory? *Journal of the Royal Society of Medicine*, 114(5): 235–236, 2021. doi: 10.1177/01410768211013525.

David Lang, Lief Esbenshade, and Robb Willer. Did ohio's vaccine lottery increase vaccination rates? a pre-registered, synthetic control study. *Journal of Experimental Political Science*, 10(2):242–260, 2023. doi: 10.1017/XPS.2021.32.

Jongkwan Lee and Hee-Seung Yang. Pandemic and employment: evidence from covid-19 in south korea. *Journal of Asian economics*, 78:101432, 2022.

Timo Mitze, Reinhold Kosfeld, Johannes Rode, and Klaus Wälde. Face masks considerably reduce covid-19 cases in germany. *Proceedings of the National Academy of Sciences*, 117 (51):32294–32301, 2020. doi: 10.1073/pnas.2015954117.

Christopher Robertson, K Aleks Schaefer, and Daniel Scheitrum. Are vaccine lotteries worth the money? *Economics Letters*, 209:110097, 2021.

Kate Rogers. Restaurants grapple with vaccine mandate policies in historically tight labor market, 2021. URL https://tinyurl.com/yemwnsgr. Published by CNBC, August 27, 2021. Accessed 2025-08-20.

Mor Saban, Vicki Myers, Shani Ben Shetrit, and Rachel Wilf-Miron. Issues surrounding incentives and penalties for covid-19 vaccination: The israeli experience. *Preventive Medicine*, 153:106763, 2021. ISSN 0091-7435. doi: https://doi.org/10.1016/j.ypmed.2021.106763. URL https://www.sciencedirect.com/science/article/pii/S0091743521003327.

Serena Tinari and Catherine Riva. Donuts, drugs, booze, and guns: what governments are offering people to take covid-19 vaccines. *BMJ*, 374, 2021. doi: 10.1136/bmj.n1737. URL https://www.bmj.com/content/374/bmj.n1737.

Robert Towey. New york city mayor bill de blasio will mandate covid vaccines for certain indoor activities, 2021. URL https://tinyurl.com/yh8hasxz. Published by CNBC, August 3, 2021. Accessed 2025-08-20.

Allan J. Walkey, Anica Law, and Nicholas A. Bosch. Lottery-based incentive in ohio and covid-19 vaccination rates. *JAMA*, 326(8):766–767, 08 2021. doi: 10.1001/jama.2021.11048.

Xiaoxuan Yang. Does city lockdown prevent the spread of covid-19? new evidence from the synthetic control method. *Global Health Research and Policy*, 6:1–14, 2021.